Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

R Fundamental and Statistical Analysis for Beginners

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Conducted by

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Version 15.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	Legacy release	Legacy master trainer slides and original R exercises.	Tertiary Infotech Academy Pte Ltd
15.0	20 August 2026	Complete source-aligned rebuild: visual deck, detailed Learner Guide, slide-mapped Lesson Plan, twelve R labs with Excel workbooks, and aligned WA/Case Study assessments.	Tertiary Infotech Academy Pte Ltd

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Course Overview

This 2-day, 16-hour course develops beginner-to-intermediate capability in R fundamentals, programming, visualisation and statistical analysis through twelve progressive labs.

Learning Outcomes

- LO1: Install R and RStudio IDE and work confidently with scripts, the console and help system.
- LO2: Select, create, inspect and manipulate R data types and data structures.
- LO3: Use R packages and datasets and import, validate and export external data.
- LO4: Create and interpret appropriate data visualisations in R.
- LO5: Write R programs using conditions, loops and reusable functions.
- LO6: Apply descriptive statistics, correlation, regression, hypothesis testing and ANOVA in R.

Daily Schedule

Time	Duration	Topic / Activity	Method	Slides
Day 1 · 09:00–09:45	45 min	Administration, outcomes and assessment briefing	Visual briefing / discussion	Slides 1–14
09:45–11:00	75 min	Topic 1 · Getting Started in R · Labs 1–2	Demonstration / guided lab	Slides 15–30
11:00–11:10	10 min	Morning break	Break	—
11:10–13:00	110 min	Topic 2 · Data Types · Labs 3–4	Visual teaching / guided lab	Slides 31–60
13:00–13:45	45 min	Lunch	Break	—
13:45–15:20	95 min	Topic 3 · Packages and Data I/O · Labs 5–6	Demonstration / guided lab	Slides 61–76
15:20–15:30	10 min	Afternoon break	Break	—
15:30–18:00	150 min	Topic 4 · Data Visualisation · Labs 7–8	Visual teaching / lab / reflection	Slides 77–96
Day 2 · 09:00–11:10	130 min	Topic 5 · R Programming · Labs 9–10	Demonstration / guided lab	Slides 97–114
11:10–11:20	10 min	Morning break	Break	—
11:20–13:00	100 min	Topic 6 · Statistical Analysis · Lab 11	Visual teaching / guided lab	Slides 115–141
13:00–13:45	45 min	Lunch	Break	—
13:45–15:30	105 min	Topic 6 · Hypothesis testing and ANOVA · Lab 12	Visual teaching / capstone lab	Slides 142–144
15:30–15:40	10 min	Course summary and assessment preparation	Review	Final summary slides

Time	Duration	Topic / Activity	Method	Slides
15:40–16:40	60 min	Written Assessment (SAQ)	Individual open-book assessment	—
16:40–18:00	80 min	Case Study	Individual open-book assessment	—

Topic-by-Topic Delivery

Topic 1: Getting Started in R · Divider 15 · Slides 16–30

Coverage: R · RStudio · scripts · comments · help. Mapping: K4 · A2.

- Teach and interpret: What is R? — A script turns an interactive calculation into an auditable analysis.
- Teach and interpret: Install R and RStudio — Verify the engine before troubleshooting packages or scripts.
- Teach and interpret: RStudio Interface — Use the Source pane as the record of analysis; use the Console as feedback.
- Teach and interpret: R Scripts and Comments — Readable scripts are easier to review, rerun and repair.
- Teach and interpret: Help and Self-Diagnosis — Good R users do not memorise everything; they diagnose efficiently.

Lab 1: Your First Reproducible R Script · Slides 26–27

Facilitate from the detailed Learner Guide and lab README. Verify: Script runs from a clean R session without manual Console steps.; Summary excludes the missing value and reports the expected valid count.; Evidence sheet records the output filename and verification result.

Lab 2: Help, Objects and Session Diagnostics · Slides 28–29

Facilitate from the detailed Learner Guide and lab README. Verify: Diagnostics identify the missing argument or wrong object type.; Corrected expression returns the expected value.; Session information is captured for reproducibility.

Topic 2: Data Types · Divider 31 · Slides 32–60

Coverage: atomic values · vectors · matrices · arrays · data frames · lists · factors. Mapping: K3 · K6 · K7 · A5 · A9.

- Teach and interpret: Atomic Values and Variables — Type is a property of the object, not the variable name.
- Teach and interpret: Character Data — Text cleaning should preserve meaning while making categories consistent.
- Teach and interpret: Vectors — Think in complete columns rather than one cell at a time.
- Teach and interpret: Indexing and Missing Values — Filtering is part of the analysis logic and must be documented.
- Teach and interpret: Vector Statistics and Arithmetic — Compare location and spread before making a claim about typical values.
- Teach and interpret: Matrices — Always state the dimension and fill direction when constructing a matrix.
- Teach and interpret: Arrays — Name dimensions whenever an axis carries business meaning.
- Teach and interpret: Data Frames — Inspect column names and types before filtering or modelling.

- Teach and interpret: Filtering and Subsetting — A filter is a business rule expressed as code.
- Teach and interpret: Lists — Lists let one analysis result carry data, model and diagnostics together.
- Teach and interpret: Factors — Factor levels encode meaning, not just display order.
- Teach and interpret: Choosing the Right Structure — Data structure is an analytical design decision.

Lab 3: Vectors, Indexing and Missing Values · Slides 56–57

Facilitate from the detailed Learner Guide and lab README. Verify: No invalid or missing reading enters the final mean.; Mean, median and IQR match the expected values.; All filtering rules are visible in the R script.

Lab 4: Data Structures and Factors · Slides 58–59

Facilitate from the detailed Learner Guide and lab README. Verify: Matrix contains only numeric values and has the expected dimensions.; Data frame preserves column types.; Factor levels and counts are explicit and correct.

Topic 3: R Packages and Data I/O · Divider 61 · Slides 62–76

Coverage: packages · built-in datasets · working directories · CSV/Excel · validation · export. Mapping: K3 · K6 · A2 · A5 · A9.

- Teach and interpret: Packages and Libraries — Separate environment setup from analysis execution.
- Teach and interpret: Built-in Datasets — Metadata is part of the dataset.
- Teach and interpret: Working Directory and Paths — Portable paths are essential for reproducible course labs.
- Teach and interpret: Import and Validate Data — Validation protects every downstream chart and model.
- Teach and interpret: Export and Audit — An export should be traceable back to its input and script.

Lab 5: Packages and Built-in Datasets · Slides 72–73

Facilitate from the detailed Learner Guide and lab README. Verify: Script uses requireNamespace() before optional installation.; Data profile reports correct dimensions.; Package and R versions are recorded.

Lab 6: Import, Validate and Export Weather Data · Slides 74–75

Facilitate from the detailed Learner Guide and lab README. Verify: Validation stops the script if a required column is absent.; May average uses only non-missing ozone records.; Export contains source row count and analysis timestamp.

Topic 4: Data Visualisation · Divider 77 · Slides 78–96

Coverage: scatter · line · boxplot · bar · pie · histogram · interpretation. Mapping: K3 · K5 · A3 · A4 · A7 · A10.

- Teach and interpret: Chart Selection — State the question first, then choose the chart.
- Teach and interpret: Scatter and Line Plots — Read direction, form, strength and unusual points.
- Teach and interpret: Multiple Scatter Plots — Exploration can be broad; communication should be selective.
- Teach and interpret: Boxplots — Compare centre, spread, overlap and unusual observations.
- Teach and interpret: Bar Charts — Bar length is precise; decorative 3D effects are not.
- Teach and interpret: Pie Charts — Use a pie only when the part-to-whole message matters.
- Teach and interpret: Histograms — Read centre, spread, skew, modes and unusual gaps.

Lab 7: Base R Chart Selection · Slides 92–93

Facilitate from the detailed Learner Guide and lab README. Verify: Each chart answers a distinct question.; Axes, units and categories are readable.; Interpretations describe evidence without claiming causation.

Lab 8: Compare Distributions with Boxplots · Slides 94–95

Facilitate from the detailed Learner Guide and lab README. Verify: Boxplot includes every feed group.; Summary table agrees with the plotted medians and spread.; Interpretation distinguishes observation from causal claim.

Topic 5: R Programming · Divider 97 · Slides 98–114

Coverage: conditions · loops · next/break · operators · functions · arguments. Mapping: K6 · K7 · A7 · A9.

- Teach and interpret: Conditions — A condition converts a rule into an explicit decision.
- Teach and interpret: While Loops — Every while loop needs a clear exit path.
- Teach and interpret: For Loops — Clarity matters more than avoiding loops at all costs.
- Teach and interpret: next and break — Explain why records are skipped or processing stops.
- Teach and interpret: Operators — Readable expressions reduce logical defects.
- Teach and interpret: Functions and Arguments — A small, focused function is easier to test than repeated code.

Lab 9: Conditions, Loops, next and break · Slides 110–111

Facilitate from the detailed Learner Guide and lab README. Verify: Missing values are skipped and counted.; Safety stop is explicit and testable.; Loop and vectorised count agree before the stop point.

Lab 10: Build and Test Reusable Functions · Slides 112–113

Facilitate from the detailed Learner Guide and lab README. Verify: Invalid n or sides values stop with clear messages.; Default and named argument calls both work.; Tests are reproducible under the recorded seed.

Topic 6: Statistical Analysis with R · Divider 115 · Slides 116–144

Coverage: descriptive statistics · correlation · regression · tests · ANOVA · diagnostics. Mapping: K1 · K2 · K5 · A1 · A3 · A4 · A6 · A7 · A8.

- Teach and interpret: Descriptive Statistics — Describe the data before modelling it.
- Teach and interpret: Trimmed Mean and Robustness — Robustness is a deliberate trade-off between sensitivity and stability.
- Teach and interpret: Correlation — A negative r means one variable tends to decrease as the other increases.
- Teach and interpret: Linear Regression — The slope quantifies the expected change in y for one unit of x.
- Teach and interpret: Residuals and Mean Squared Error — Residual patterns reveal model problems that one score can hide.
- Teach and interpret: Regression Diagnostics — Diagnose unintended outcomes before trusting a model.
- Teach and interpret: Multiple Regression — More predictors do not automatically mean a better model.
- Teach and interpret: R-squared and Model Fit — Model fit is evidence, not a guarantee of accuracy.
- Teach and interpret: Hypothesis Testing — Predefine the question and threshold before looking at the result.

- Teach and interpret: Two-Sample t-Test — Report the group estimates, confidence interval and p-value.
- Teach and interpret: One-Sample t-Test — Pair the p-value with the estimated difference and interval.
- Teach and interpret: ANOVA — ANOVA answers an overall question; post-hoc analysis answers which groups differ.

Lab 11: Descriptive Statistics, Correlation and Regression · Slides 140–141

Facilitate from the detailed Learner Guide and lab README. Verify: Correlation interpretation includes direction and strength.; Model formula and prediction target are explicit.; Diagnostic evidence is included before a conclusion.

Lab 12: Hypothesis Testing and ANOVA Capstone · Slides 142–143

Facilitate from the detailed Learner Guide and lab README. Verify: Every conclusion uses reject/fail-to-reject language correctly.; ANOVA conclusion does not overstate which groups differ.; The updated script reruns successfully and the audit note explains the change.

Resources Required

- R 4.x and RStudio Desktop
- Microsoft Excel or LibreOffice
- readxl R package
- Course slides, Learner Guide and all twelve lab folders
- LMS access and SSG digital attendance access

Assessment

Written Assessment (SAQ) — 7 open-ended questions, K1–K7, 60 minutes, open book.

Case Study (CS) — 6 open-ended tasks, A1–A10, 80 minutes, open book.

The assessment is individual and open book. Candidate papers are uploaded to the LMS. A minimum of 75% digital attendance and a Competent result are required for funding outcomes.